# 2. Methods for Measuring and Evaluating Health Hazards

Assume for a moment that you have been called to a local plant to act as an occupational safety and health consultant. The plant manager tells you that he has only recently become aware of the OSHA law and feels that his plant may be in violation of that law in some areas. He would like you to study the plant for potential health hazard exposures, identify those exposures that exist, and make recommendations concerning controls that can be effected to remove these health hazard exposures. In this situation, what would you do? How would you proceed to identify the problems? Obviously you cannot go helter-skelter throughout the plant looking here and there in hopes that you will identify a health hazard exposure. As a consultant, your daily rate is high enough that you must provide the plant manager with a maximum return for his dollar invested. Therefore, it is imperative that you organize your approach in such a manner that the results obtained will be the best possible.

Although the same general considerations should exist if you were hired as a full-time employee on the staff of the plant, it may not be the case. The individual who works as an employee of an organization may not be asked to justify his salary and the results obtained for that salary as is the case with the consultant. Consequently, the employee may be caught up in the act of being a member of the "fire brigade," putting out "little fires" as they occur rather than attempting to organize and plan the work in such a manner as to show major accomplishments. It is suggested that whether one is a consultant or an employee of a plant, one should attempt to obtain the maximum result for each dollar expanded.

In the field of occupational safety and health, it is desirable that, rather than reacting to problems, the professional plan and organize his work to seek out problems before they become emergencies that need immediate solutions. This is not always going to work; but as a general rule, most emergencies could be prevented with proper planning.

The purpose of this discussion is to present an organized method for recognizing and evaluating health hazards within the occupational environment. The method that is presented is a suggested organization method and is not the only way this can be done. However, the basic principles employed here should be present in any occupational safety and health study. As a matter of fact, the general principles presented here are applicable to other professions as well. Only the specific details would be changed.

## A Suggested Analysis Study Outline

Your overall objective as an industrial hygiene engineer is to recognize potential hazard exposures, determine if they are in fact really exposures, and develop methods for controlling those hazard exposures that do exist. This seems simple enough. However, consider the case study presented at the end of this chapter. In the case study, you are to assume the position of a consultant responsible for occupational health and safety within the facility. On further consideration, the job is not so simple as it seemed at first glance. Many potential hazards exist. Many real hazards may be present. Alternate methods for control are available to be considered. Where do you start and what do you do to attempt to bring some order to the situation? In the following pages, a suggested method for accomplishing your job in a reasonable amount of time and with a reasonable amount of effort is presented.

### Recognition of Hazards

Walk-Through Survey. Perhaps the first thing you might do on completing initial discussions with the plant manager is to conduct a walk-through survey of the plant. It is of value to carry a portable tape recorder with you to make notes on things that you see during this survey. The purpose of such a survey is not to identify all hazard exposures that may exist or to identify solutions that might be used to control hazards that you may feel exist but rather to gain a general feel for the operation. Your first objective is to become familiar with the overall plant, its products, its people, its general layout, and the work environment as a whole. Notes concerning these can be made for later reference. Remember, this is only an initial walk-through survey; it may be necessary to go back and look at things in more detail later.

Preliminary Hazard Recognition. After becoming familiar with the overall operation of the facility, you should make a more detailed study of the plant to identify potential hazard exposures. To do this, you will need your notes from the initial walk-through survey, layouts of the individual plant areas and the plant as a whole, process flows, process descriptions, job descriptions, and procedures in use. Now organize your thoughts in terms of a given plant area: for example, metal fabricating, vapor degreasing, paint shop, etc. Each of these areas must be studied in detail using the layouts, process flows, and other information that pertains to it.

Beginning with one of the areas chosen, consider the types of exposures that may exist: for example, airborne contaminants, physical hazards, surface-contacting materials, flammable materials, etc. By considering the types of exposures that a worker in the area might face, you will be less likely to miss potential hazards. In each of these types of exposures, consider where a possible potential hazard exposure to the worker may exist within the plant area. This can be done by reviewing your notes, the process, job descriptions, and procedures that are available. The emphasis is on identifying all potential hazard exposures, not just those that appear to exist. In this way it is possible to include exposures that may be present but easily overlooked.

In reviewing the operation for the various types of exposures, tools such as fault-tree analysis, job-safety analysis, and failure mode and effect are of value. These methods were discussed in the previous chapter. Through the use of these tools, it is possible to describe where potential problems mayexist that are not necessarily obvious. The industrial hygiene engineer should become familiar with the use of these and similar tools to improve the results obtained from a preliminary hazard analysis.

The preliminary hazard worksheet (Figure 1.2.1) presents a method for listing the results of your initial survey. For each type of exposure, a description of the potential hazard is given. This description should contain an explanation of what the potential hazard is, the probable source of the hazard, and the possible cause. The description must of necessity be brief; however, it should be concise enough to pinpoint the problem being considered.

The next important item to consider is the number of employees potentially affected by the hazard. This information will help to identify those hazards that require the most immediate action. Obviously all hazards should be controlled, but this may not be feasible given the time required to accomplish the control. The important thing is to solve first things first and go from there in an organized manner.

It is important to note the methods that are currently being used to control the hazard exposure. Just because controls exist, one cannot assume that the exposure is being controlled. Quite often malfunction of a control system can create a hazard exposure more serious than that which would exist in the absence of any control. If a control exists, it should be carefully evaluated to assure that it is working properly.

The next item to consider for a given hazard exposure is its level of effect on the human system. For purposes of this discussion, this level of effect has been broken down into three categories: a low level, where the hazard is of annoyance to the workers and has possible harmful effects; a medium effect, including those hazards that are dangerous to the health of humans; and a high effect, including those hazards that cause death or severe injury to the worker's health. This classification system is only suggested. Other classification systems can be developed that may be more useful. The major use of such a classification is to provide criteria upon which the decision concerning the priority of action can be made.

One further determination is useful before deciding the priority of study for each of the hazard exposures identified. It is desirable that the professional make some judgment concerning the probability that the hazard does, in fact, exist. In many cases, it is possible to determine that there is little likelihood that the exposure is not being controlled even though the potential hazard is a very serious one. For example, the absence of layers of dust in an area that would be expected to have a high concentration of airborne particles certainly indicates that the controls being used are relatively effective. Other factors such as experience and the presence or absence of worker complaints can help the professional to determine whether there is a high likelihood that a problem exists.

Figure 1.2.1

## PRELIMINARY HAZARD RECOGNITION WORKSHEET

Date _____

Area _____ Investigator _____

| TYPE OF EXPOSURE | DESCRIPTION OF POTENTIAL HAZARD | EMPLOYEES POTENTIALLY AFFECTED | PRESENT CONTROLS | LEVEL OF EFFECT LOW MED HIGH | PROBABILITY OF HAZARD EXISTING LOW MED HIGH | PRIORITY |
|---|---|---|---|---|---|---|
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Finally, after all potential hazard exposures in each given area have been identified and described, it is desirable to determine those exposures that should be studied and in what order they should be studied. To determine the order in which problems will be studied, a subjective priority should be placed on each hazard as it relates to the other potential hazards. This priority is determined by considering both the level of effect and the probability of existence for each exposure. Those hazard exposures with a high level of effect and a high probability of existence should be studied first, while those potential hazard exposures with a low level of effect and a low probability of existence can be considered only after the more likely and dangerous exposures have been controlled.

Upon completion of preliminary hazard recognition, the industrial hygiene engineer has a list of hazard exposures that can serve as a basis for planning and organizing his work for a long time in the future. Such a method can also provide a justification for neglect of a particular problem at a given time while others may feel that this problem should be under study. Without such a system, it is likely that the industrial hygiene engineer will be forced to react to every request for study that comes up, since there is no way to compare this hazard to other potential hazards that also require study.

## Hazard Assessment Study Design

Preliminary Hazard Assessment. The first step in a hazard assessment study design is to make a preliminary determination concerning the possible extent of the hazard. If such a determination was made during the preliminary hazard recognition stage, it will not require repetition at this point. If such a determination was not made, it should be, since quite often a complete study of potential hazard exposure requires a significant expenditure of time and money. If preliminary results indicate that the problem is not significant, then this expenditure of time and money can be avoided.

Various alternatives exist to accomplish a preliminary hazard assessment, depending on the type of exposure being assessed. For example, if an air quality problem is being studied, colorimetric indicators can be used to obtain initial results. If the function of a ventilation system is in question, smoke tubes might be used to gain a quick estimate concerning the capture velocity of the system. Sound level readings might be made at key spots within an area to determine if a noise problem appears to be significant, or the industrial hygiene engineer might assess the difficulty in conversation within an area to gain a quick estimate concerning the level of noise that is present. These and similar techniques can provide an initial screening of the problem to determine whether further study is required, thus saving valuable time and money.

Hazard Assessment Study. If the results of the preliminary hazard assessment seem to indicate the need for further study, then a complete hazard assessment study must be conducted. The specific methods to be used will depend on the problem being faced. There are, however, certain general characteristics of a hazard study design that should be considered by the industrial hygiene engineer.

The first consideration is: Has the hazard exposure been defined completely? Do you know what type of hazard exposure may exist? Where is the likely source of the exposure? In what form will the hazard be found? This information should be available from the preliminary hazard recognition, but it is worthwhile to reconsider the problem definition at this point to be sure that the study will obtain the information necessary to determine whether or not the hazard exposure is present.

A second major consideration is choosing between alternative methods for obtaining desired results. Quite often the tendency is to design a study without considering possible alternative designs that might be used. Some time should be taken to assure that alternative designs do not exist that will provide better results or are easier to implement.

Related to the above consideration is the selection of equipment to be used to determine the hazard level. There are various types of measuring instruments available that can be used in a hazard assessment study. Each of these types of equipment has advantages and disadvantages and is best suited for certain types of assessment studies. The description of the hazard and the results to be obtained must be considered in choosing the appropriate equipment to use. Again alternatives should be considered to assure that the best equipment is used. One restraint on this choice of alternatives is the equipment that is available within the plant. Although gas chromatography may be the best method for analyzing a given air sample, this equipment may not be available for use either within the plant itself or in the local area. Thus, it may be necessary to choose an alternate method to obtain the results.

The definition of the problem will help define the kind of equipment that is required. For example, in an air quality study, the sampling train that is to be used may be set up to provide instantaneous samples or integrated samples, depending on the desired results. This decision must be considered, given the definition of the problem.

A clear definition of the target limit that the study results should not exceed is required. Such a definition in terms of the units to be considered will assist in defining the equipment and sampling methods to be used. Is the desired result in terms of peak levels, or is it in terms of a time-weighted average concentration of the hazard? What is the upper limit to be considered as safe? This may be the published threshold limit value, or it may be some other figure, depending on the existence of a threshold limit value or other requirements. This should be clearly defined at the outset of the study so that the decision criteria for further action are known.

When designing the study, consider the possibility of installing continuous monitoring equipment if the hazard is extremely dangerous to the worker and could get out of control quickly. Such a continuous monitoring system can be used to determine the levels present during the study and can remain in effect to assure that the limit is not exceeded in the future.

Another important consideration in designing a hazard assessment study is the location at which the sampling is to take place. Again this depends on a definition of the problem that may be faced. Where are the workers within the

potential hazard area? Are they stationary, or do they move from area to area? The use of dosimeters can provide useful information concerning workers who move about. A stationary sampling point is certainly adequate when the worker does not move about the work area. It is also important to consider the method of entry of the sample into the human system. If the pollutant is likely to enter the human system as a result of respiration, the sampling point should be at the breathing zone level, either on the individual worker or stationary at this level within the area.

With the exception of a continuous sampling device, the time at which samples are taken is important. Hazard exposures may vary according to the time of day, and this variation should be considered when setting up the sampling plan. Quite often a buildup over a particular shift will occur, or the buildup may occur in relation to processing cycles. If this type of situation is expected, it is desirable to obtain multiple samples during the various expected cycles to determine if such a buildup is occurring. Another point to consider in terms of time is that, although the operation may be shut down during a third shift, there may be requirements for maintenance workers to operate in the area at this time. The exposure of maintenance workers should also be considered, and samples may be required during this third shift.

The next step in designing a hazard assessment study is to develop a step-by-step procedure that identifies what is to be done to assess the hazard exposure. This step-by-step procedure should include any evaluation steps and define any calculations that are necessary. This is done to assure that the study design plan is well thought out before entering into the study itself. If the study has not been well designed, it is possible that a number of false starts will occur, each requiring unnecessary expenditure of time and money; or that the results obtained will not be adequate to make the decision. Then the study will have to be repeated using different procedures. The procedure being used may be of a branching nature so that, given a certain occurrence, certain calculations or procedural steps must be performed; while given another occurrence, other steps or calculations will be performed. It is also possible that interim procedural steps may indicate that the study is no longer necessary. This type of branching can save time and money in accomplishing the study.

One important consideration in a hazard assessment study is the documentation of the study and its results. This documentation can provide information that will be useful in the future should other studies be required. The documentation can also justify the study and the results obtained. If a question arises concerning the results, it will be easy to identify exactly what was done and how the results were obtained. A sample for documenting the study design is given in Figure 1.2.2. A study outline is shown in Figure 1.2.3.

## Summary

The preceding discussion has been presented to outline a procedural method that can be used to recognize and evaluate hazard exposures that may be present within the work environment. The objective of this discussion was to provide a logical method of approach that can be used for any hazard

exposure. Obviously, certain types of exposures require that specific methods be used to assess their level. Regardless of these specific methods, it is necessary to approach the problem in a systematic manner.

It is not within the scope of this text to review the various types of measurement and evaluation techniques that are available and that can be used for a given hazard exposure. To do so would require much more space than is available. This subject matter has been covered well in other documents. Within the later chapters of this text, certain specific measurement equipment will be considered for the particular type of exposure being discussed. For example, in the area of ventilation, the various devices that can be used to measure and evaluate the effects of a ventilation system will be discussed in some detail.

Figure 1.2.2

Hazard assessment design.

---

Area _____ Type of Exposure _____

Date _____

Potential Hazard Description _____

_____

_____

Number of Employees Potentially Exposed _____

Desired Level To Be Maintained (TLV) _____

Results Obtained from Study _____

Action Recommended _____

_____

PRELIMINARY HAZARD ASSESSMENT

Equipment _____

Procedure _____

_____

Result/Recommendation _____

_____

_____

_____

_____

_____

_____

Figure 1.2.2 (Continued)

Equipment _____

_____

Location of Sampling Points _____

_____

Time Schedule for Sampling _____

_____

Procedure _____

_____

_____

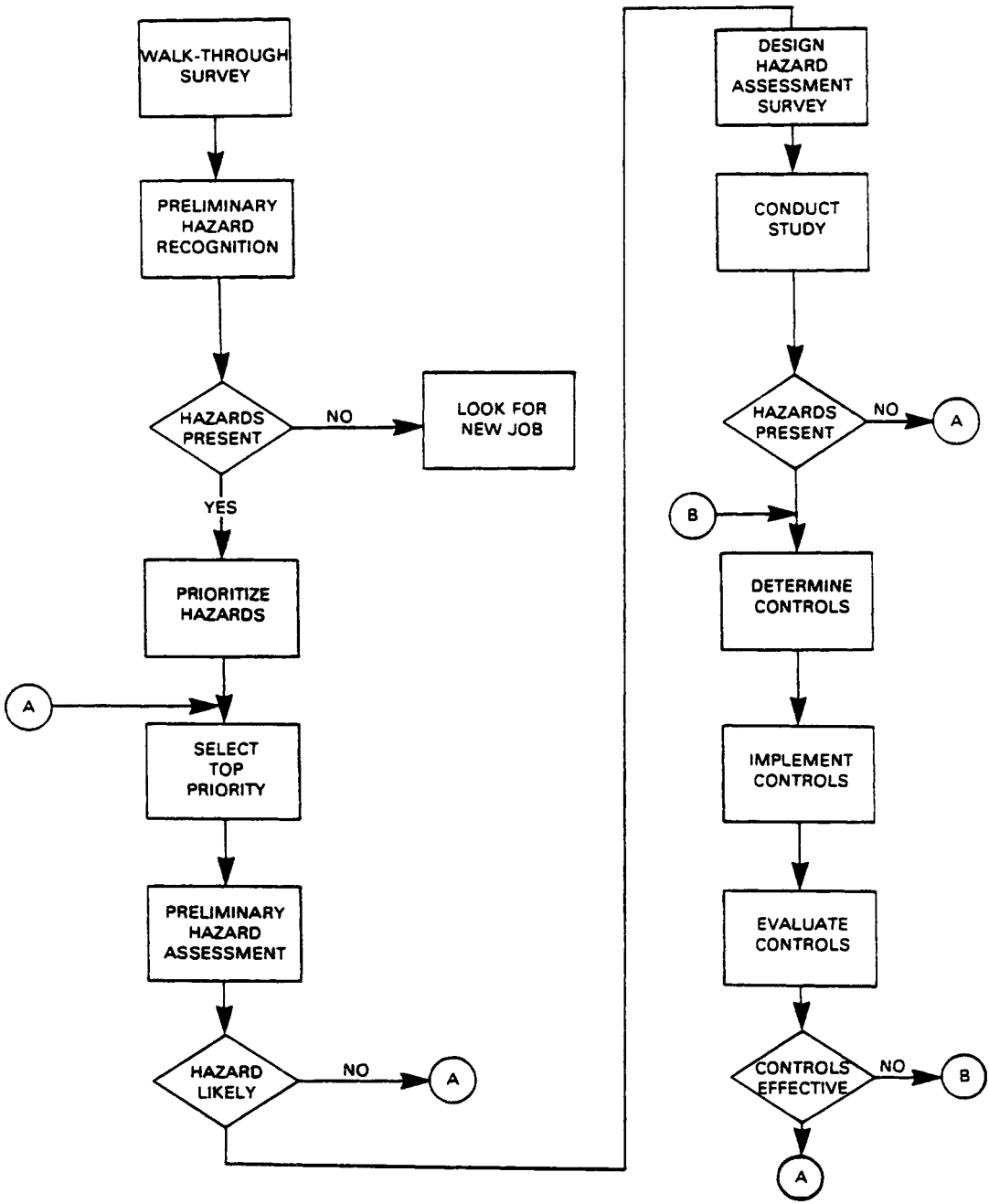
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Figure 1.2.3

Hazard assessment study outline.



## CASE STUDY

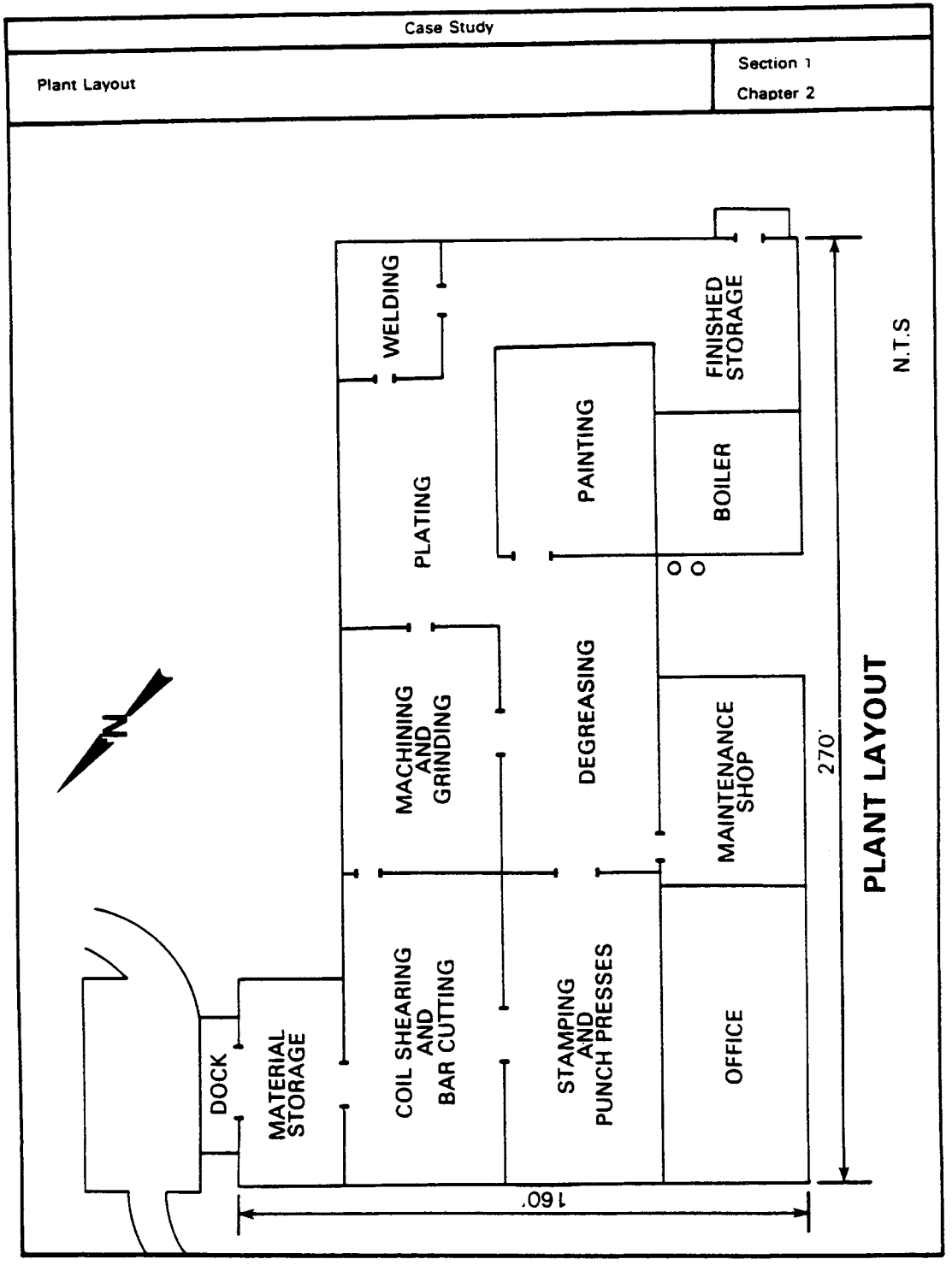
### Introduction

The following pages describe a case concerning an industrial firm, Acme Metal Fabricating, Inc. (AMF). This case may be used for study throughout the text. The case attempts to describe a realistic industrial plant that faces many occupational health and safety problems. The reader may chose to return to this case study and analyze it in an attempt to make his learning experiences more realistic.

### General Description

Acme Metal Fabricating, Inc. (AMF) is a small metal fabricating firm operating on the west side of Centertown, Ohio (population 16,500). AMF employs 94 people including office and design engineering staff. The plant is a large, single-story building with dimensions of 270 ft. by 120 ft. and open ceiling heights of 16 ft. in the production and storage areas. Office ceilings are 10 ft. AMF fabricates large and small metal parts used by other manufacturers in the local area.

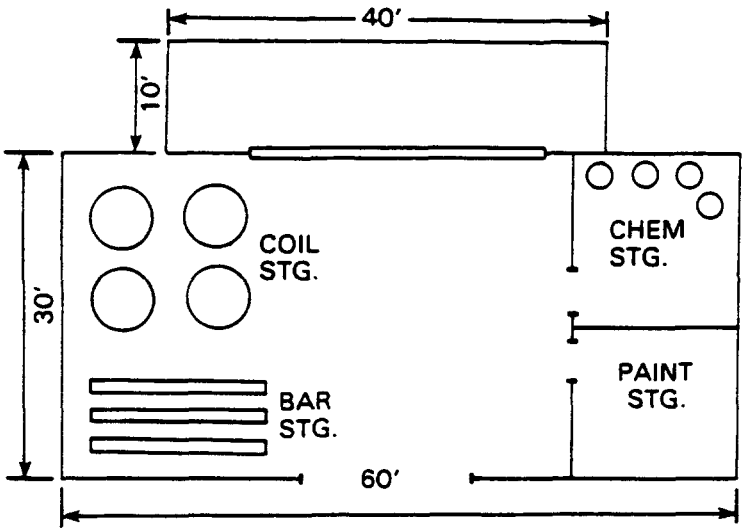
The Plant Manager, Jim Brown, has recently become aware of the possible existence of conditions that are hazardous to the health and safety of the workers. Having no one who is responsible for the health and safety of the workers, he has contacted your firm, Health and Safety, Inc., for the purposes of determining if hazards are present and, if so, recommending controls to alleviate the hazards. You and some other members of the firm's staff have been chosen to act as consultants to AMF. Mr. Brown has provided the following descriptive information concerning the operation of AMF to your group.


Case Study
Plant Layout
Section 1
Chapter 2
N
WELDING
PLATING
PAINTING
FINISHED STORAGE
BOILER
MACHINING AND GRINDING
DEGREASING
MAINTENANCE SHOP
DOCK
MATERIAL STORAGE
COIL SHEARING AND BAR CUTTING
STAMPING AND PUNCH PRESSES
OFFICE
160'
270'
PLANT LAYOUT
N.T.S

Case Study

Material Storage

Section 1
Chapter 2




N.T.S.

PERSONNEL

2 FORKLIFT DRVRS.
1 WAREHOUSE WKR.
1 RECEIVING CLRK.

## Process Description

Steel Coils

1. Steel coil is delivered by truck.
2. Forklift truck (LPG) moves coils from truck to storage. Coils weigh from 8,000# to 15,000# each depending on steel gauge.
3. Coils are stored on end.

Bar Stock

1. Bar stock is delivered by truck.
2. Forklift (LPG) moves bar stock to storage racks. Bars may vary in size from 1" x 2" x 6' to 6" x 8" x 6'.

Chemicals and Welding Materials

1. Chemicals delivered by truck in 55-gal drums. Acetylene, carbon dioxide, LPG, and oxygen are delivered in cylinders.
2. Forklift truck moves chemicals and cylinders to storage room door.
3. Drums and cylinders are moved manually using a hand cart which is pushed to storage area and tilted to slide drum or cylinder off into storage.
4. Drums are stored on end. Cylinders are stored horizontally on the floor.
5. Forklift trucks are charged with LPG in the area outside the chemical storage room.

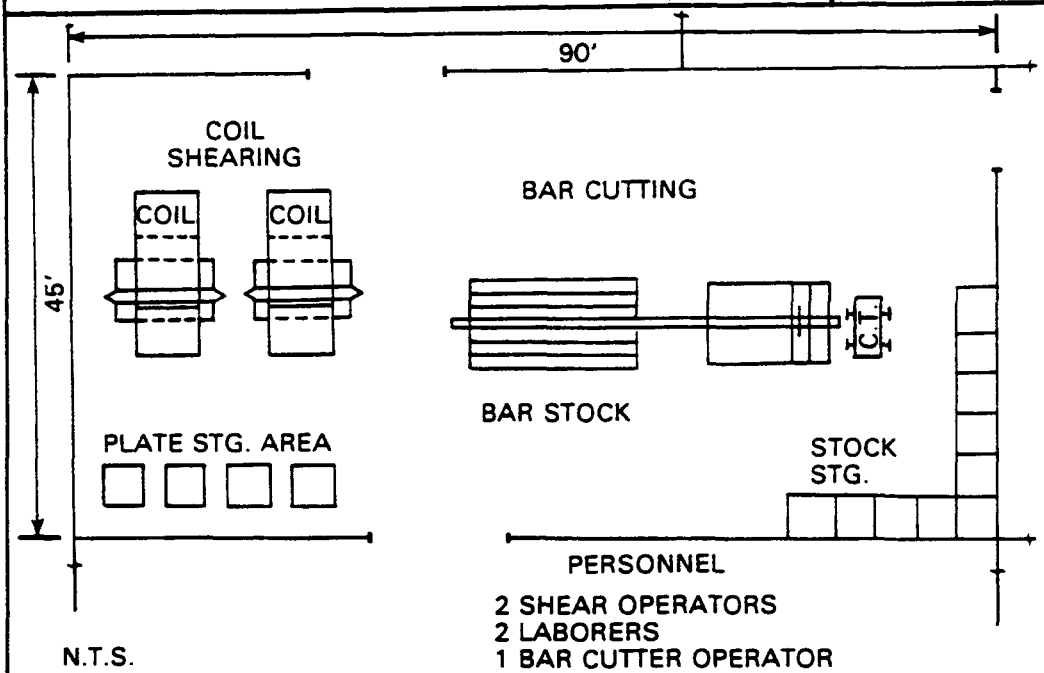
Paint

1. Paint is delivered by truck on a pallet in 5-gal cans.
2. Forklift truck moves the pallet to storage room door.
3. Cans are manually carried to a 3-level rack and placed in labeled positions.

Case Study

Coil Shearing and Bar Cutting Shop

Section 1
Chapter 2




Process Description

Coil Shearing

1. Coil is moved from storage to the shears by forklift truck (LPG).
2. Forklift truck loads shear with coil.
3. If coil size needs changing, forklift truck assists in move. Used coil is returned to storage.
4. Steel coil is sheared to required size and automatically stacked on pallet.
5. Forklift truck moves pallets from shear end to in-process plate storage.

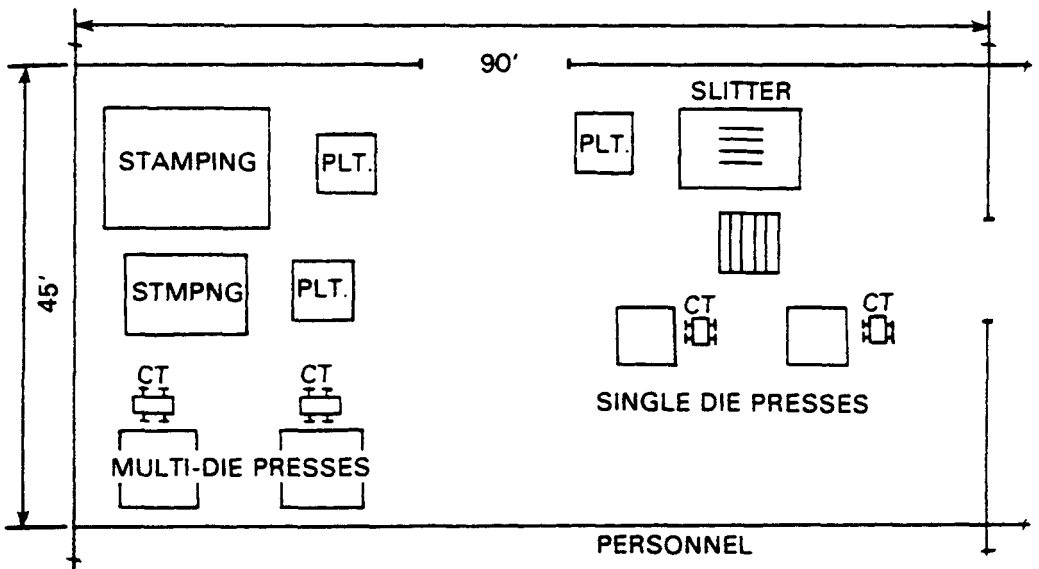
Bar Cutting

1. Stock is moved from storage by forklift truck to in-process bar storage.
2. Overhead manual crane is used to position bar for cutting shear.
3. Bar stock is cut to specification for the individual places to be machined.
4. Finished stock is manually moved by cart to finished stock storage rack.

Case Study

Stamping and Punch Press Shop

Section 1
Chapter 2




Process Description

Stamping

1. Plate is delivered by forklift truck to stamping machine.
2. Single plate is automatically fed by vacuum to stamping area.
3. Operator activates stamping mechanism approximately once every 1-2 minutes.
4. Stamped part is placed in truck wagon by operator who manually removes the part from the stamping machine.
5. Operator changes dies and press punch to meet specifications.

Slitting

1. Plate is moved from storage to slitting operation by forklift truck.
2. Plate is slit to specified size by manually moving plate through slitters.
3. Finished slit plate is manually stacked for punching.

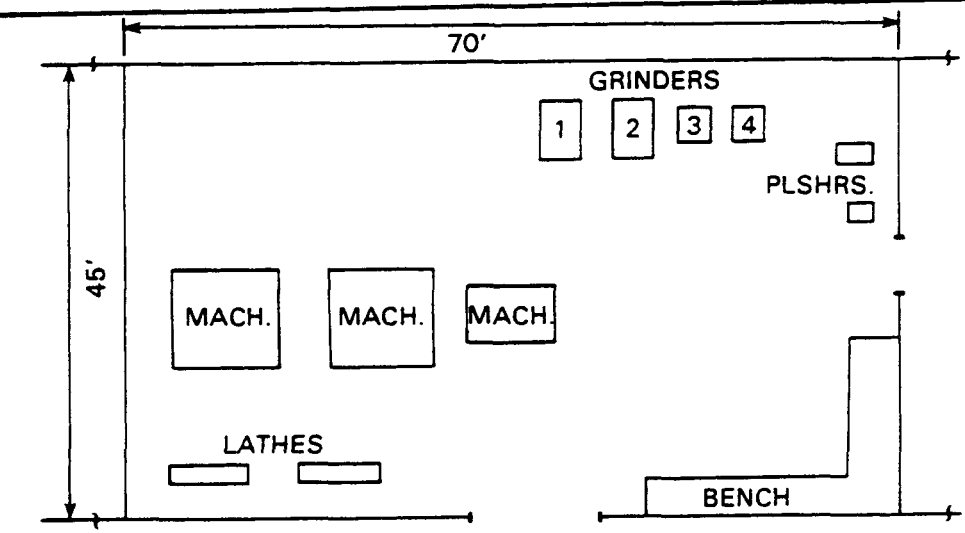
Punch Presses

1. Two single die punch presses are used for punching split plate. Two multi-die presses are used for plate punching.
2. Plate is moved to multi-die presses by forklift truck. Slit plate is manually loaded in single die presses.
3. All presses operate automatically with operator feeding stock and removing finished product from stacker.
4. Finished products are stored in tote boxes in small cart that can be pulled manually or by forklift.
5. Operator changes punches and dies to meet specifications.

Case Study

Machining and Grinding Shop

Section 1
Chapter 2




N.T.S.

Process Description

Machining

1. Cut bar stock is moved to milling machines by cart pulled manually or by forklift.
2. Cut bar stock is machined to specification using metal milling machines. Cutting oils are used to assist in the milling process.
3. Bar stock is turned on metal lathe to specification using cutting oils to assist in the process.
4. Finished parts are moved to grinders, polishers, or to degreasing by cart pulled manually or by forklift.

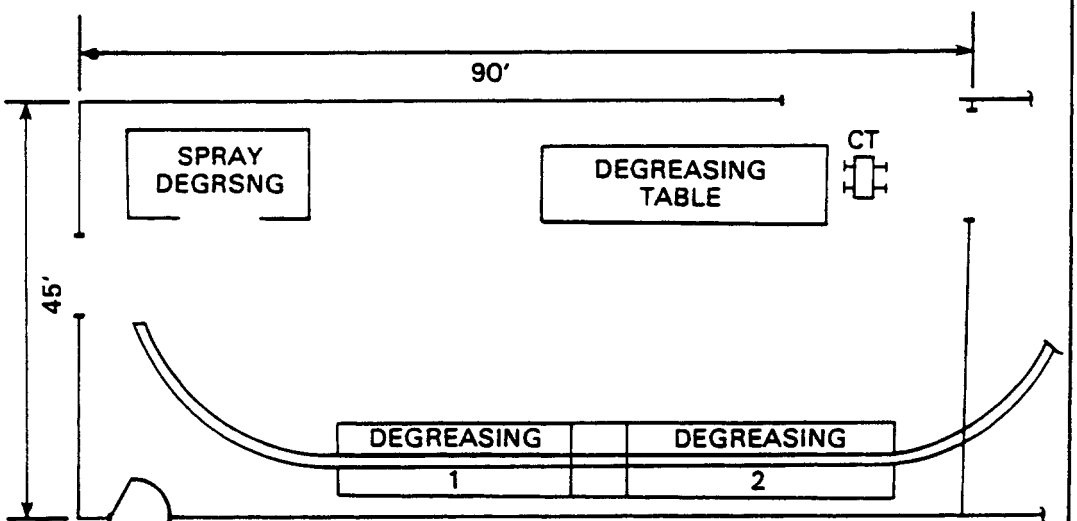
Grinding and Polishing

1. 2-3" grinders 18" in diameter operating at 150 RPM. 2-2" grinders 12" in diameter operating at 180 RPM. 1-4" polisher 18" in diameter operating at 250 RPM. 1-3" polisher 12" in diameter operating at 350 RPM.
2. Parts are ground and polished to meet specifications.
3. Completed parts are moved by hand cart to work bench for inspection and hand finishing.
4. Hand finishing is done using small 1" grinding and polishing wheels of 4" diameter operating at 550 RPM.

Case Study

Degreasing Shop

Section 1
Chapter 2




PERSONNEL

1 DEGREASER OPR.
3 MANUAL DEGREASERS
1 SPRAY DEGREASER
2 LABORERS

N.T.S.

Process Description

Large Parts

1. Large parts moved by forklift truck pulling cart to vapor degreaser.
2. Degreaser workers remove parts from wagon and attach to overhead conveyor.
3. Degreaser operator operates controls to move part into degreaser. Operator is responsible for timing part in degreaser.
4. Degreaser consists of 2-tank process. Part immersed in first tank; vapor bathed in second tank.
5. Tanks are steam heated to 190°. Solvent used is trichloroethylene with stabilizers added. Thermostat controls vapor line.
6. Tanks are cleaned and charged every 48 hours of operation.
7. Each tank is 23' long, 5' wide, and 10' high. Solvent is 4' deep.

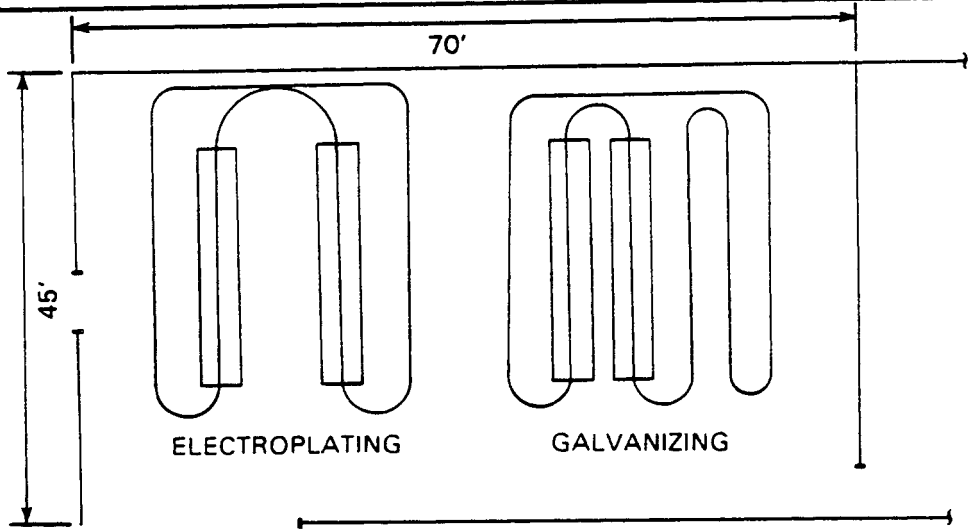
Small Parts

1. Small parts are delivered from either machining/grinding or the punch press operation by cart to degreasing tables and booth.
2. At degreasing table small parts are attached to hand operated conveyor chain and dipped into small tank of trichloroethylene at room temperature, removed, brushed with paint brush to loosen oil and grease, and allowed to dry while hanging on chain conveyor.
3. Other small parts are attached to chain in a small spray booth. Operator sprays parts to remove oil and grease and moves to drying position to air dry. Small hand spray is used to apply solvent.
4. Dried parts put in cart for painting, electroplating, or welding as required.

Case Study

Electroplating Shop

Section 1
Chapter 2




PERSONNEL

2 GALVANIZING WKRS. 2 LABORERS
2 CHROME PLTRS.
2 INSPECTORS

N.T.S.

Process Description

Chrome Plating

1. Parts are delivered by hand cart to the overhead conveyor.
2. Parts are attached to overhead conveyor, which is a manually operated motor-driven system, to move parts through the plating process.
3. A pre-plating acid cleaning process is accomplished by dipping the parts into a solution of sulfuric acid.
4. The chrome plating dip tank is filled with chromic acid with additives to reduce misting.
5. Chrome plated parts are inspected while on the conveyor.
6. Parts are removed from the conveyor to hand carts.

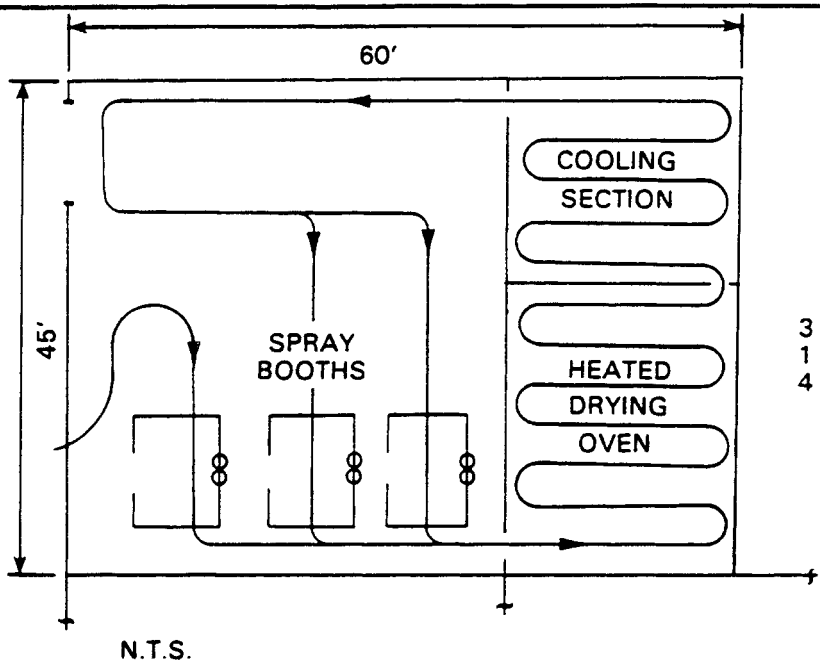
Galvanizing

1. Parts are delivered by hand cart to the overhead conveyor.
2. Parts are attached to the overhead conveyor, which is a manually operated motor-driven system, to move parts through the plating process.
3. A pre-plating acid cleaning process is accomplished by dipping the parts into a solution of sulfuric acid.
4. Galvanizing is accomplished by dipping parts into the molten zinc tank.
5. Galvanized parts are inspected on the conveyor.
6. Parts are removed from the conveyor to hand carts.

Case Study

Paint Shop

Section 1
Chapter 2

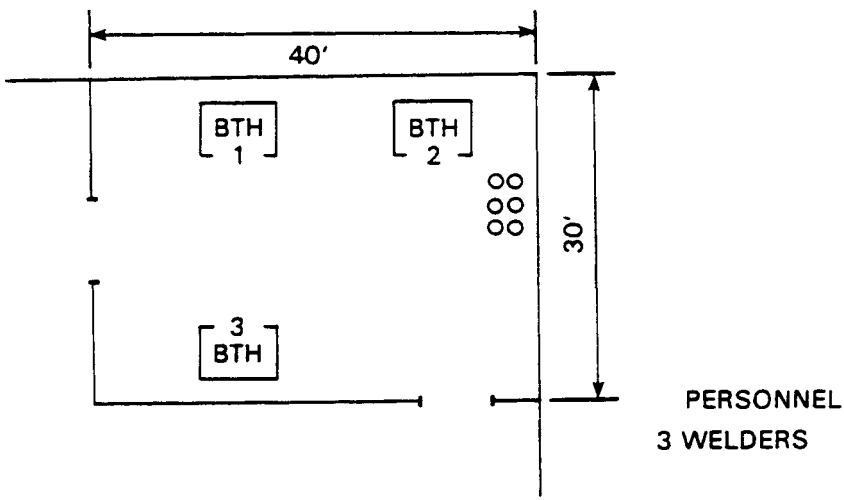



Process Description

1. Small parts are delivered by hand cart and attached to one of two parallel overhead conveyors. Large parts that are to be painted are delivered to the spray booth conveyor. Parts not painted are removed from conveyor in paint shop and carried by hand cart to electroplating, welding, or storage as required.
2. Painter sprays parts using hand-held spray gun.
3. Paint used is leaded zinc oxide base paint.
4. Parts moved from spray booth to a pickup conveyor.
5. Parts are moved continuously through drying oven operating at 300°F.
6. Final stage of drying is at cool temperature, 60°-70°F.
7. Parts are removed from conveyor and placed on cart equipped with racks to be moved to storage by forklift truck.

Case Study

| Welding Shop | Section 1<br>Chapter 2 |
|---|---|




N.T.S.

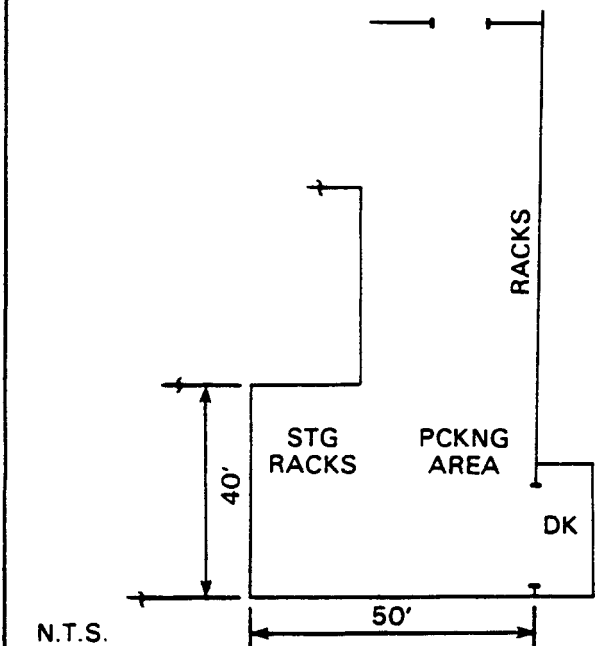
Process Description

1. Parts are delivered to the welding shop by hand truck.
2. Welding is done to specification on cleaned parts.
3. Some of the galvanized parts are sent to welding.
4. Booths #1 and #2 are used for acetylene cutting and welding using an oxygen assist.
5. Booth #3 is used for gas shielded metal arc welding. Carbon dioxide is used as the gas shield.
6. Parts are moved from welding by cart pulled manually or by fork-lift.

Case Study

Finished Storage

Section 1
Chapter 2




PERSONNEL
3 PACKERS
1 FORKLIFT DVR.
2 WAREHSMEN
1 SHIPPING CLK.

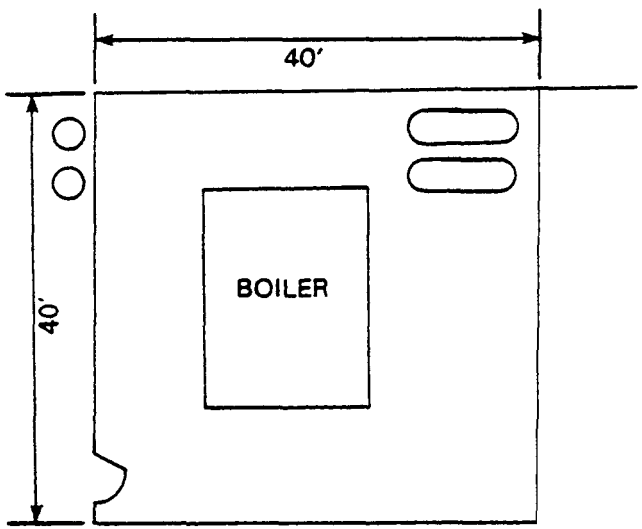
Process Description

1. Parts are moved to storage racks by cart pulled manually or by forklift truck.
2. Parts are stored by type on storage racks. Small parts are in tote boxes. Large parts are on pallets.
3. Storage racks are 12' high.
4. Parts are delivered to the packing area by forklift truck or by cart pulled manually.
5. Parts are manually packed in boxes and palletized.
6. Pallets are loaded into trucks by forklift truck.

Case Study

Boiler Room

Section 1
Chapter 2




PERSONNEL
2 BOILER OPRS.

N.T.S.

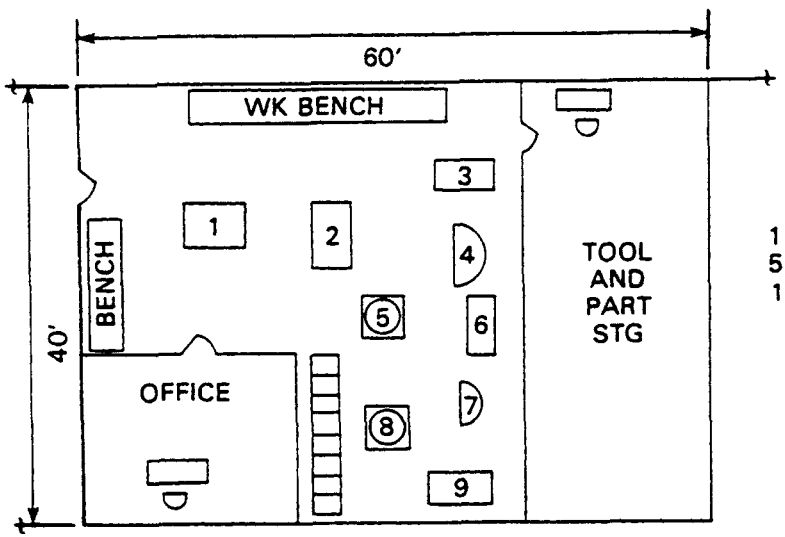
Process Description

1. Boiler is gas fired with a capacity of 750,000 lbs of steam/hour at 2000 psi and 1000°F.
2. One stack is used for boiler flue gas exhaust.
3. The second stack is used for ventilation of gases gathered from the Degreasing and Paint shops.
4. Two feedwater treatment tanks are located in the boiler room.

Case Study

Maintenance and Machine Shop

Section 1
Chapter 2




PERSONNEL
1 MAINT. SUPVSR.
5 MAINT. WORKERS
1 TOOL CLERK

N.T.S.

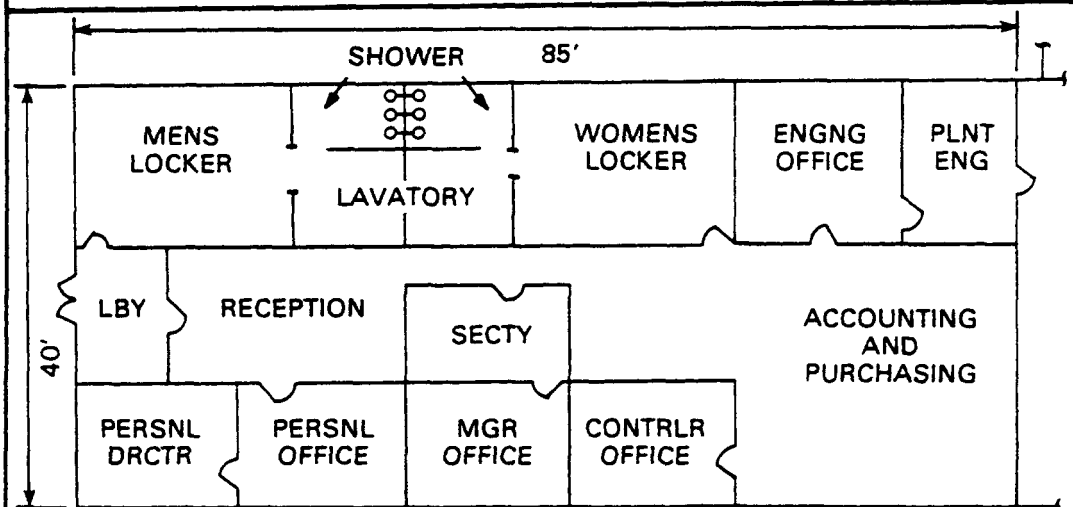
Process Description

1. Miscellaneous small parts required for maintenance operations are fabricated here.
2. Equipment includes
   a. wood working
      1. table saw
      2. jointer
      3. sander
      4. band saw
      5. drill press
   b. metal working
      6. lathe
      7. band saw
      8. drill press
      9. pipe cutter
3. The Tool and Part Storage area is adjacent to the maintenance facility.

Case Study

General Office

Section 1
Chapter 2




PERSONNEL

| | |
|---|---|
| 1 PLANT MGR | 1 CONTROLLER |
| 1 PERSNL DRCTR | 6 ACCTG AND PRCHNG |
| 1 GENERAL FRMAN | 3 DRAFTSMEN |
| 1 PLANT ENGINEER | 5 SECRETARIES |

N.T.S.

Process Description

1. The General Office area is a single story area which is air conditioned.
2. Wall between the office area and plant is exterior brick covered by lath and plaster since this area was added to the plant 10 years after the plant was built (3 years ago).